ALICE L WALTON SCHOOL OF MEDICINE

Listeria monocytogenes

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Adapted from: Listeria monocytogenes

Listen to this Brick
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Learning Objectives (4) Versions

After completing this brick, you will be able to:

1. List the taxonomy of *L. monocytogenes.*
2. Compare and contrast the diseases and epidemiology of *L. monocytogenes.*
3. Explain how the virulence factors of *L. monocytogenes* contribute to disease progression.
4. Interpret labo

CASE CONNECTION

AW, a 73-year-old man, comes to your office with nausea, vomiting, and diarrhea. "This started yesterday," he reports. "Right now, the main issue is my stomach. I've already thrown up four times today." He has no other symptoms. You broaden your history to inquire about travel, change in dietary habits, antibiotic use, and exposure to anyone who has had a similar illness. "I was visiting my son's farm. I did drink a lot of milk. I don't usually drink much milk anymore, but he has it fresh there, and it was really delicious. He says it's from the grass the cows eat. Did I drink too much milk?" On exam, AW's abdomen has generalized mild tenderness. There is no blood in the stool.

How did AW become ill? Consider your answer as you read, and we'll revisit at the end of the brick.

GO TO CONCLUSION ↓

Taxonomy: Listeria monocytogenes? (LO1,4)

Listeria monocytogenes can be easily identified in the laboratory based on its unique characteristics. *L. monocytogenes* exhibits growth across a broad range of temperatures, pH values, and salt content, which enables the bacteria to persist even in cooked, refrigerated foods.

L. monocytogenes bacteria are gram-positive, facultative intracellular rods that tend to grow in clust pattern of clusters or pairs is similar to *Streptococcus*. Unlike most bacteria, *L. monocytogenes* likes colder temperatures—in fact, these bacteria grow well in culture in temperatures as low as 4°C (39.2°F), known as "cold enrichment."

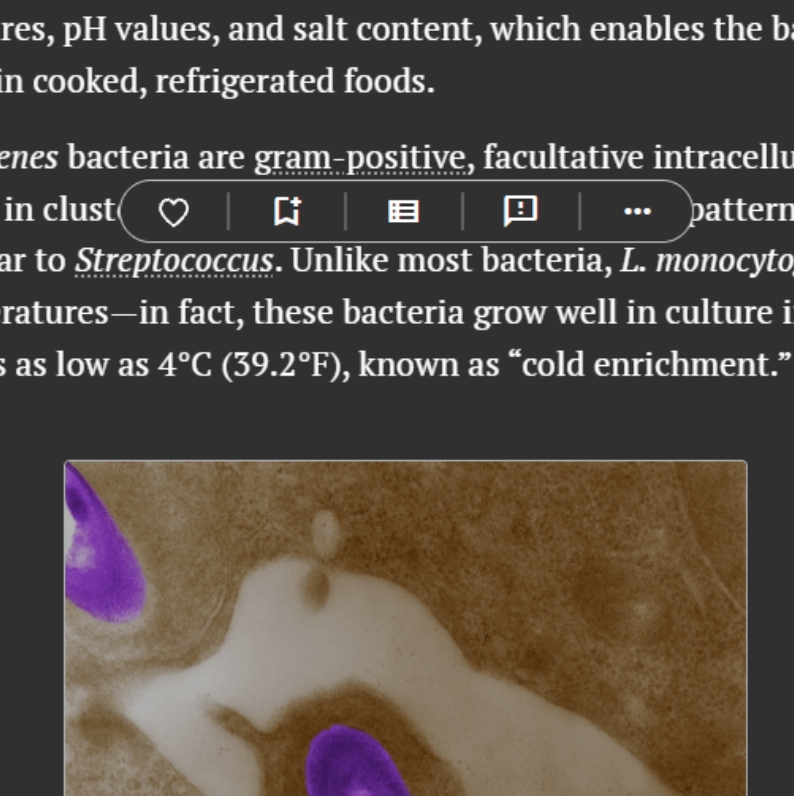


Figure 1

Credit: Courtesy of the CDC/Dr. Balasubr Swaminathan and Peggy Hayes.

Mnemonic

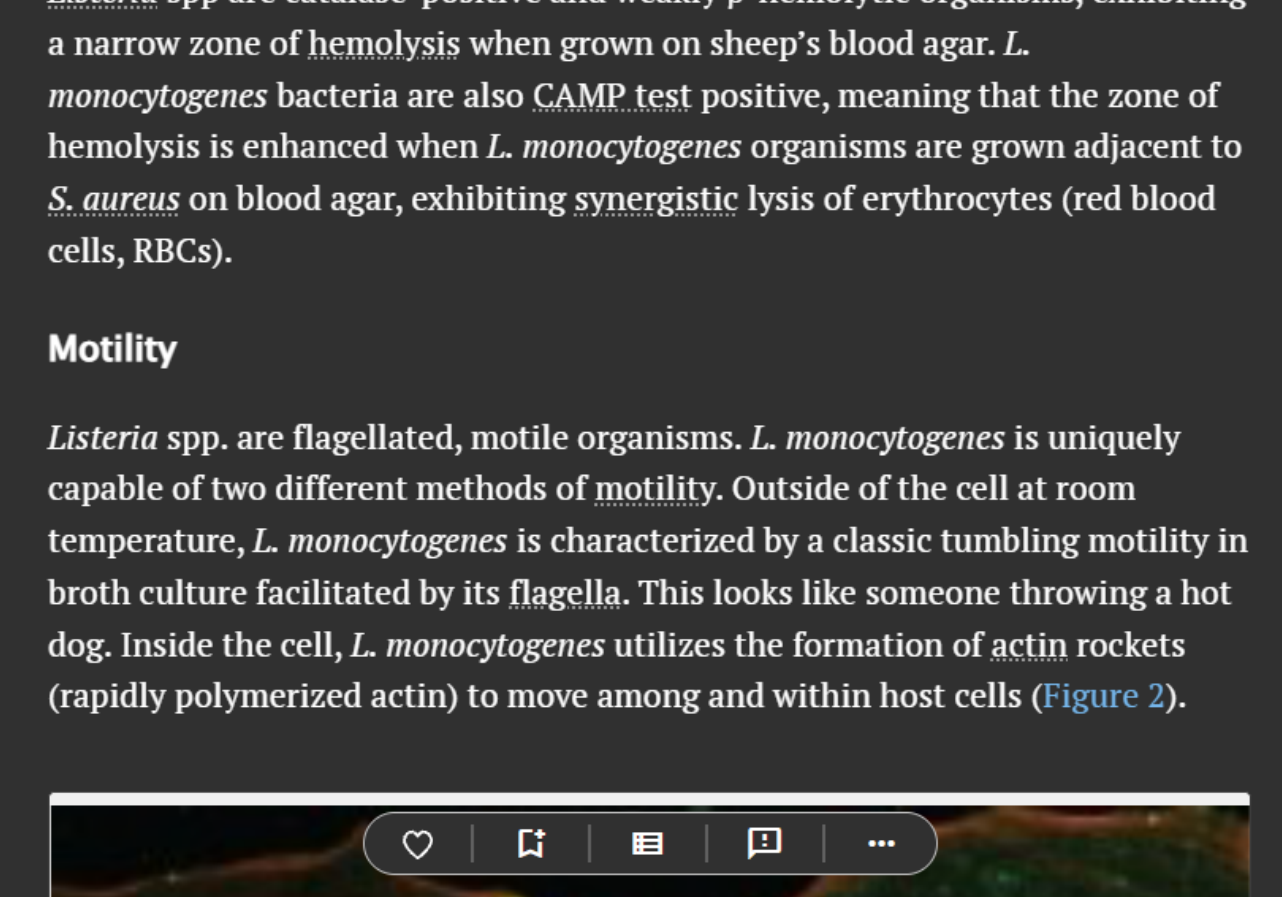
Listeria monocytogenes grows in **cool** temperatures, so think **Cool Mint *Lister*mean.**

Biochemical and Metabolic Characteristics

Listeria spp are catalase-positive and weakly β-hemolytic organisms, exhibiting a narrow zone of hemolysis when grown on sheep's blood agar. *L. monocytogenes* bacteria are also CAMP test positive, meaning that the zone of hemolysis is enhanced when *L. monocytogenes* organisms are grown adjacent to *S. aureus* on blood agar, exhibiting synergistic lysis of erythrocytes (red blood cells, RBCs).

Motility

Listeria spp. are flagellated, motile organisms. *L. monocytogenes* is uniquely capable of two different methods of motility. Outside of the cell at room temperature, *L. monocytogenes* is characterized by a classic tumbling motility in broth culture facilitated by its flagella. This looks like someone throwing a hot dog. Inside the cell, *L. monocytogenes* utilizes the formation of actin rockets (rapidly polymerized actin) to move among and within host cells (Figure 2).



QUIZ

Figure 2

Credit: Courtesy of Schuppler M et al. Int J Inflamm 2010;2010:704321.

What type of hibit at room temperature?

Epidemiology: L. monocytogenes (LO2)

L. monocytogenes is abundant in the environment and can be found in plant life, soil, water, and in animal carriers, including humans. It can also be isolated from animal feces. Humans may carry a low level of *L. monocytogenes* as gastrointestinal flora. *L. monocytogenes* infections typically only cause severe disease in a few patient populations—neonates, pregnant people, the elderly, and the immunocompromised (particularly those with defective cellular immunity) are especially at risk of invasive disease. Certain occupations, including butchers, farmers, veterinarians, slaughterhouse workers, poultry farmers, and fish mongers at higher risk due to increased exposure. Healthy adults and children are susceptible to milder disease presentations.

Which patient po *. monocytogenes* infection?

The most common route of transmission is via ingestion of contaminated foods. Unpasteurized dairy products, soft cheeses, raw vegetables, and processed meats are possible sources. *L. monocytogenes* has been identified as the causative organism in several foodborne disease outbreaks, with food items such as peaches, cantaloupes, deli meats, and bean sprouts subject to recall due to reports of listeriosis. *L. monocytogenes* can also be acquired via vertical transmission. Pregnant people are 10 times more likely to get listeriosis than the general population.

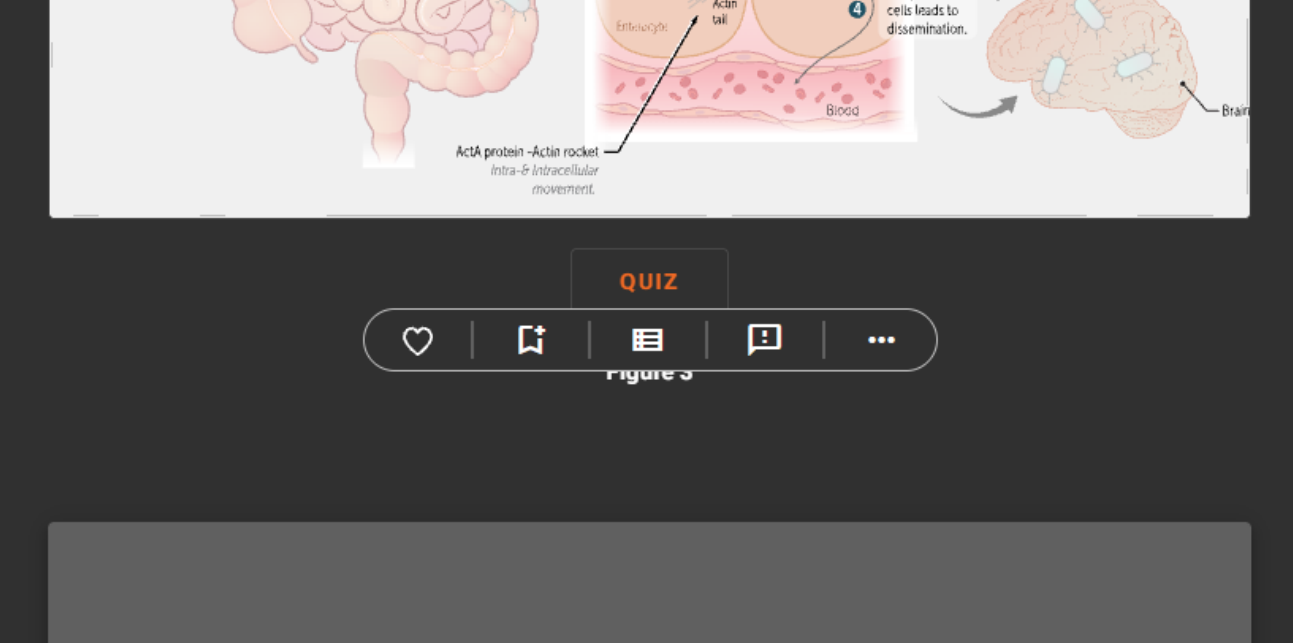
CLINICAL CORRELATION

In 2024, the CDC reported a multistate outbreak of listeriosis that infected 34 people. Interviews revealed that the majority of patients had eaten meats sliced at deli counters. Boar's Head recalled 7 million pounds of deli meat due to contamination concerns. Every year, roughly 1,600 people in the U.S. contract listeriosis, and 260 die from the illness.

Clinical presen s: L. monocytogenes (LO2,3)

L. monocytogenes bacteria are equipped with several clinically relevant toxins and virulence factors that contribute to disease in humans (Figure 3). They produce:

- **Listeriolysin O and phospholipase C enzymes:** Help bacteria avoid digestion in phagosomes following host cell entry via phagocytosis; mechanism of this action leads to host cell membrane disruption, such as hemolysis of erythrocytes
- **ActA:** Protein that mediates intracellular motility via actin polymerization


Listeria monocytogenes
1 Digestion of contaminated food/water.
Adhesion and invasion 2 of GI epithelium.
InlA or InlB
Internalin surface proteins
InlA: Enterocytes
InlB: Endothelial, fibroblasts, enterocytes etc.
Listeriolysin O & Phospolipase C
Evade phagosomes
Actin polymerization and propulsion through cytosol (ActA) 3
Enterocyte
Actin tail
4 Entry into circulating cells leads to dissemination.
Blood
Brain
ActA protein -Actin rocket
Intra-& Intracellular movement.

QUIZ

Figure 3

Which virulence factor produced by virulent strains of *L. monocytogenes* is associated with the coordination of actin movement?

Neonates

L. monocytogenes can be transmitted transplacentally. If infection occurs early in the pregnancy, it may cause spontaneous abortion, still birth, or premature birth. A severe form of early-onset disease in the newborn may present as disseminated granulomas and abscesses in multiple organs (granulomatosis infantiseptica). Late-onset disease in a neonate presents after birth and is characterized by meningitis, meningoencephalitis, or septicemia.

Pregnant People

Cellular immunity in e third trimester, which is when these patients are most susceptible to *L. monocytogenes* infection. Typically, the resulting flu-like illness is mild, nonspecific and self-limiting. Symptoms include fever, myalgias, and arthralgias.

Healthy Adults

Healthy adults may acquire a transient *L. monocytogenes* infection that leads to listeriosis, which is associated with the consumption of contaminated food and is characterized as a febrile gastroenteritis with diarrhea. This gastroenteritis is usually mild and self-limited in healthy nonpregnant adults.

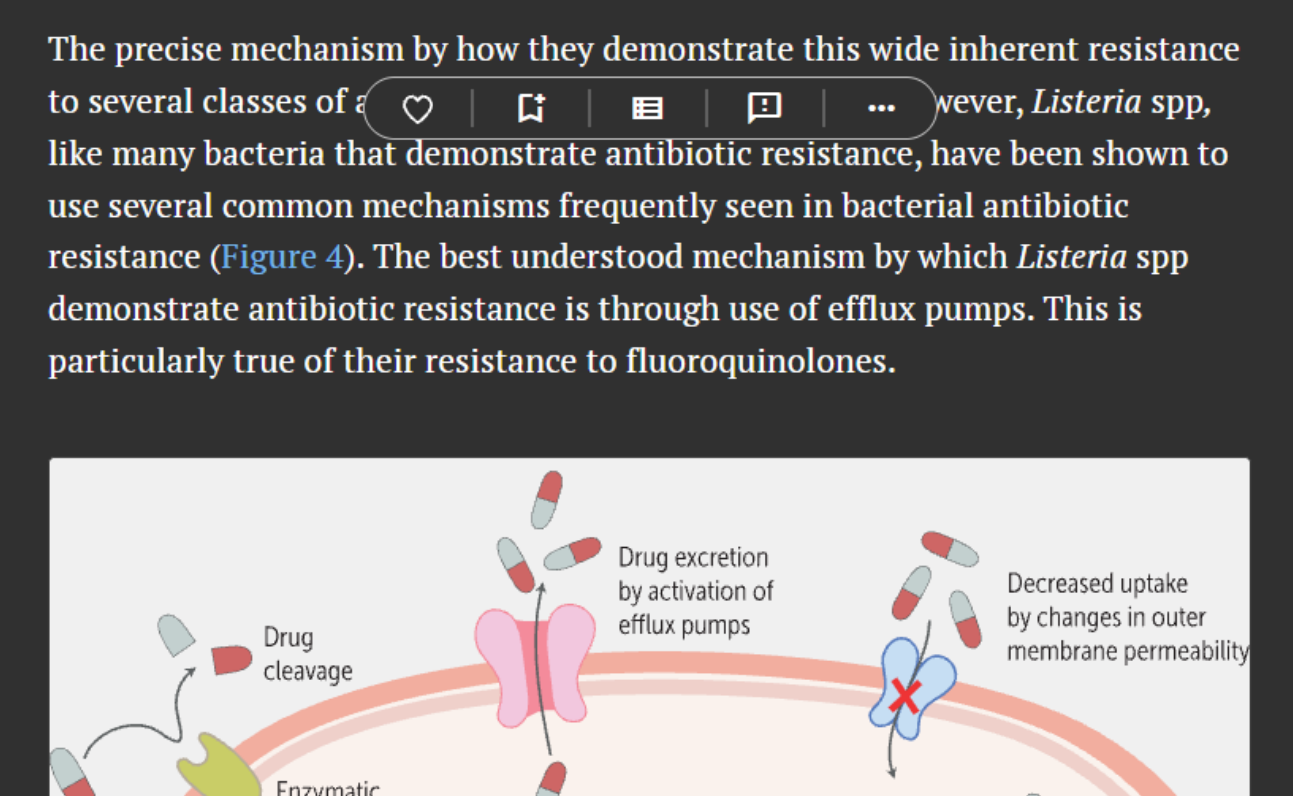
Elderly Patients and Patients with Defects in Cellular Immunity

Patients with defects in cellular immunity include those with acquired immune deficiency syndrome (AIDS), lymphomas, or a history of transplant. Listeriosis in these patients, and in the elderly, is more severe than in healthy adults. *L. monocytogenes* is also the most common cause of meningitis in transplant patients.

L. monocytogenes Antibiotic Resistance

Because *L. monocytogenes* is a common foodborne pathogen that can cause serious illness or, in some cases, prove fatal, the development of antibiotic resistance in this species is of great concern. *L. monocytogenes* is inherently resistant to several classes of antibiotics, including fluoroquinolones, cephalosporins, macrolides, and tetracycline.

The precise mechanism by how they demonstrate this wide inherent resistance to several classes of a wever, *Listeria* spp, like many bacteria that demonstrate antibiotic resistance, have been shown to use several common mechanisms frequently seen in bacterial antibiotic resistance (Figure 4). The best understood mechanism by which *Listeria* spp demonstrate antibiotic resistance is through use of efflux pumps. This is particularly true of their resistance to fluoroquinolones.


Drug cleavage
Enzymatic inactivation of drugs
Drug modification
Drug excretion by activation of efflux pumps
Decreased uptake by changes in outer membrane permeability
mRNA
DNA
Modification on the drug target

QUIZ

Figure 4

One hypothesis for why *L. monocytogenes* demonstrates such a wide range of antibiotic resistance relates back to its mode of transmission and the environment these bacteria occupy outside of the body. Recall that *Listeria* spp. are frequently found in unpasteurized dairy products and processed meats. Many agriculturally raised animals are prophylactically given very high doses of antibiotics in the hopes of reducing animal food product contamination. The use of antibiotics in livestock is extremely problematic, since it consistently selects for antibiotic resistance. When bacteria from livestock contaminate produce, they are commonly antibiotic resistant.

Which mechanism of antibiotic resistance does *L. monocytogenes* engage to avoid destruction by the fluroquinolone class of antibiotics?

In addition, recent research has demonstrated that *L. monocytogenes* bacteria have movable genetic elements that confer increasing resistance to penicillin and ampicillin, drugs h cases of antibiotic resistance, the use of trimethoprim-sulfamethoxazole has been proven effective in treating an *L. monocytogenes* infection.

CLINICAL CORRELATION

The drug of choice for the treatment of *L. monocytogenes* is ampicillin. It may be supplemented with gentamicin, and susceptible patients are advised to avoid the consumption of milk, soft cheeses, processed meats, and raw vegetables.

CASE CONNECTION BACK TO INTRODUCTION ↑

Thinking back to AW, how did he become ill?

AW has been drinking fresh milk. You explain to AW, "While there are many possible causes of diarrhea, nausea, and vomiting, the circumstances really point toward that fresh milk you drank, which probably contained the bacteria known as *Listeria*. Routine pasteurization at commercial dairies kills it, but the milk you drank was not pasteurized, correct?" AW confirms this. You order stool cultures to prove the diagnosis and prescribe ampicillin to treat the infection along with medication to treat the nausea so AW is able to drink and stay hydrated.

Summary

- *L. monocytogenes* ellular rod that can thrive in a broad ronments, and a wide pH range.
- Tumbling motility and actin rockets are both characteristic of *L. monocytogenes* movement.
- *L. monocytogenes* is found in plant life, soil, water, and animal carriers.
- Neonates, the elderly, pregnant people, and patients with defective cellular immunity are especially at risk for *L. monocytogenes* infection.
- *L. monocytogenes* is responsible for a wide range of human disease, including meningitis, listeriosis, septicemia, and granulomatous infantiseptica.
- Efflux pumps and moveable genetic elements may contribute to the development of antibiotic resistance in *L monocytogenes*.

Review Questions

1. A 73-year-old man presents to the clinic complaining of a stiff neck, nausea, vomiting, and a decreased level of consciousness. Gram-stain of a cerebrospinal fluid (CSF) culture is inconclusive, and cold enrichment reveals bacterial colonies. Which of the following is the most likely causative agent of this patient's meningitis?

A. ○ *Cryptococcus neoformans*
B. ○ *Haemophilus influenzae* type b
C. ○ *Listeria monocytogenes*
D. ○ *Neisseria*
E. ○ *Streptococcus pneumoniae*

Explanation (requires correct answer)

2. A patient is diagnosed with listeriosis after attending a dinner party. Which of the following is the most likely source of contamination?

A. ○ Sweet potato
B. ○ Brie cheese
C. ○ Fried rice
D. ○ Pasteurized milk
E. ○ Shellfish

Explanation (requires correct answer)

3. Which of the following patient populations is least at risk for acquiring a severe *L. monocytogenes* infection?

A. ○ Elderly adults
B. ○ Healthy a
C. ○ Neonates
D. ○ Patients with AIDS
E. ○ Pregnant people

Explanation (requires correct answer)

4. Which virulence factor of *L. monocytogenes* is used to prevent destruction by phagocytes?

A. ○ Internalins
B. ○ Listeriolysin O
C. ○ ActA
D. ○ Protein M
E. ○ Coagulase

Explanation (requires correct answer)

H ?

References

- Listeria Outbreak Linked to Meats Sliced at Delis | CDC

Review Learning Objectives

Check off the objectives when you feel comfortable with the concept

01 ☐ List the taxonomy of *L. monocytogenes.*
02 ☐ Compare and contrast the diseases and epidemiology of *L. monocytogenes.*
03 ☐ Explain how the virulence factors of *L. monocytogenes* contribute to disease progression.
04 ☐ Interpret laboratory results to identify *L. monocytogenes.*

Objective Confidence: 0/4

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